Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Bovell-Benjamin, Adelia Ph.D.		
Project Title:	Food Technology Utilization		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	334-727-8717
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City:	Tuskegee	**State:**	AL
Zip Code:	36088	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	01/01/1997	**End Date:**	12/31/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Bovell-Benjamin, Adelia		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Work continued on optimizing the sweetpotato syrup. Over the past year, an improved, acceptable color was accomplished, chemical analyses (% starch, Brix), consumer testing, descriptive analysis, and rheology studies were completed. Use of hydrochloric acid was removed from the protocol for the syrup. Sweetpotato bread. Bread was designed using four levels of sweetpotato flour (SPF) and wheat flour (WWF). The appropriate formulations for the bread were developed in our Food Processing Laboratory. Chemical analyses, loaf volume, moisture content, color, and texture. Differential Scanning Calorimetry (DSC) X-ray Diffraction, Scanning Electron Microscopy (SEM) and Nuclear Magnetic Resonance (NMR)and Stress Relaxation Tests were done on the different breads. Extruded ready-to-serve extruded breakfast cereal (RTSBC). Three different RTSBC with the following combinations were developed: 100% sweetpotato flour (SPF); 75/25%SPF/wheat bran (SPFWWB); and 100% wheat bran (WWB).		

Task Description:	Quantitative descriptive analysis and consumer testing were completed over the past year and a MS thesis was completed on the work. Shelf stability of field-grown sweetpotatoes. In collaboration with the GEDI group, FOTU evaluated the proximate composition, color, texture and microbial characteristics of six sweet potato lines (J6-76, 94-95, WC-91, UNK3, TU-00-02, and Beauregard) during storage. The beginning and ending mean moisture contents were similar for all the sweetpotatoes except J6-76 and UNK3, which were significantly lower. Protein content and weight loss were significantly higher for UNK3 than the others. Beauregard had the deepest orange color, and UNK3 was the firmest sweetpotato. At the end of the study, UNK3 had the highest bacterial count. Work continued on the peanut protein meat analog over the last year and a MS thesis is pending.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Economic value and marketability of sweetpotatoes and peanuts may be increased. Development of plant-based biodegradable food packaging on Earth. On-going partnership with NASA.
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)